

Solutions :-

1) a) The dimensionless quantities in Entropy, Solid angle, speed, Mole fraction, Energy, relative Error are :-

- * Entropy
- * Solid Angle
- * Speed
- * Energy
- * Relative Error.

b) i) $76.4 \times 180.4 = 13792.16$
 $= 2 \text{ significant figures.}$

ii) $10.3 + 2.83 = 13.13$
 $= 4 \text{ significant figures.}$

2) a) Photoelectric Effect :-

* It is a phenomenon in physics where electrons are emitted from a material when it is exposed to light or EM radiation.

Mathematical Expression for photoelectric equation is given as:

$$E = hf - \phi$$

E = kinetic Energy

h = Planck's Constant

f = frequency of light

b)

$$E = \frac{hc}{\lambda}$$

$$E = W_0 + K_{\max}$$

W_0 = work function

$K_{\max}$ = Max. Kinetic Energy.

$$K_{\max} = 0$$

$$E = W_0$$

$$\frac{hc}{\lambda} = W_0$$

$$W_0 = \frac{(6.626 \times 10^{-34} \text{ J.s}) (3.00 \times 10^8 \text{ m/s})}{6800 \times 10^{-10} \text{ m}}$$

$$W_0 = \frac{1.9878 \times 10^{-25} \text{ J.m}}{6.8 \times 10^{-7}}$$

$$W_0 \approx 2.922647 \text{ J} \quad (\text{or}) \quad 2.927$$

Heisenberg: Principle :-

• Explains about why a scientist can't predict the value of two variables ~~in~~ simultaneously in the quantum mechanics.
• mathematically, it is expressed as :

$6.26 \times 10^{-34} \text{ J.s}$

$$\Delta x \cdot \Delta p \geq \frac{h}{2}$$

→ Planck's constant.

↓
Position of Particle

→ Momentum of particle

b)

$$\Delta x \cdot \Delta p \geq \frac{h}{2}$$

 ~~$\Delta x = \text{uncertainty}$~~

$$\Delta p = m \cdot \Delta v$$

 $m = \text{mass of electron}$

$$m = 9.31 \times 10^{-31} \text{ kg}$$

$$\Delta v = 300 \text{ m/s}$$

$$\Delta x \geq \frac{h}{2 \Delta p}$$

$$\Delta x \geq \frac{6.626 \times 10^{-34} \text{ J.s}}{2 \times (2.79 \times 10^{-34} \text{ kg m/s})}$$

$$\Delta x \geq \frac{6.626}{2 \times 2.79} \times 10^{-34} \text{ m}$$

$$\Delta x \geq 1.19 \times 10^{-34} \text{ m}$$

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$\therefore$ uncertainty in the position of electron is approximately $1.19 \times 10^{-34} \text{ m}$.

4) a) Node ~~function~~:

* A node refers to a point in a standing wave (such as wave function) of particle in a box) where amplitude of the wave is zero.

⇒ wave function of a particle in a box where amplitude of the wave is zero

✂

* The wave function (ψ) for a particle in a 1-D box is given by:

$$\psi(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

Here, $\psi(x)$ = wavefunction as a function of position (x),

* L is the length of the box.

* n is the quantum number ($1, 2, 3, \dots$)

* π is a mathematical constant

ii) For $x=0$, $x=\frac{1}{2}$ for $n=2$:

$$P(0 \leq x \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} |\psi(x)|^2 dx$$

$$P(0 \leq x \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} \left[\sqrt{\frac{2}{L}} \sin\left(\frac{2n\pi x}{L}\right) \right]^2 dx$$

$$P(0 \leq x \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} \frac{2}{L} \sin^2\left(\frac{2n\pi x}{L}\right) dx$$

v) a) Reversible Thermodynamic System:-

* It is an idealized concept in Thermodynamics used to analyze & understand certain process.

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In reversible process, the system undergoes changes such that the system & its surroundings can be turned into their original state.